**Università degli Studi di Messina**

Dipartimento di Scienze Matematiche e Informatiche, Scienze Fisiche e Scienze della Terra
Bachelor's Degree in Computer Science (L-31) · Curriculum Data Analysis

# Towards Investigating the Role of Artificial Intelligence in Neurological Rehabilitation



## Measuring movement from the video of VRRS sessions

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OUTLINE

# What we will see today

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## Context

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What we learned and what comes next

# 01

# Context

Why measure movement in rehabilitation?



# Rehabilitation needs objective measures

Parkinson's disease changes movement in three ways



## Bradykinesia

Slow and small movements



## Tremor

Rhythmic shaking



## Dyskinesia

Involuntary extra movements



TODAY

The therapist observes and judges by eye. Visits are infrequent, so small changes can be missed.



GOAL OF THIS THESIS

**Get objective movement measures from the video of a normal VRRS session, with no extra sensors.**

# The VRRS platform



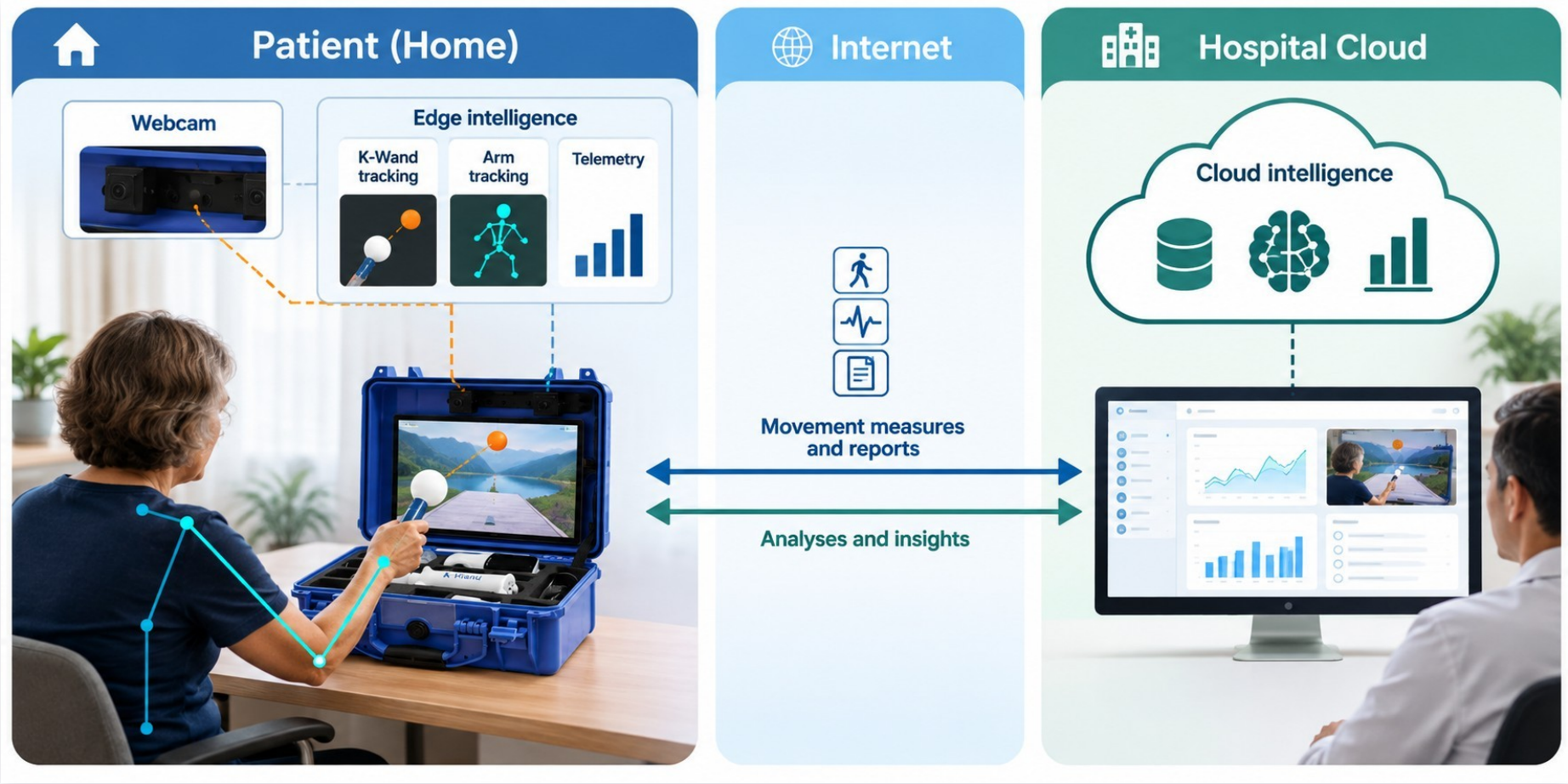

## Strengths

- Clinically established: stroke, MS, Parkinson's
- Game-like exercises keep patients engaged
- Real-time feedback from the K-Wand
- Remote sessions through telesupport

## Limits

- Dedicated hardware and infrastructure
- Needs trained personnel
- Cost-effectiveness still to be established

# The vision: from home to hospital




*In this thesis the analysis ran offline on recorded sessions; the picture shows the intended use.*

# 02

# Method

How the system works



# How the videos were recorded

### 1 Session

The patient moves the K-Wand

### 2 Telesupport

Patient, physician and game together

### 3 Recording

OBS Studio · 60 fps · 3360×2100

### 4 Analysis

Our program, offline on a laptop

### 5 Results

CSV per frame, JSON per session

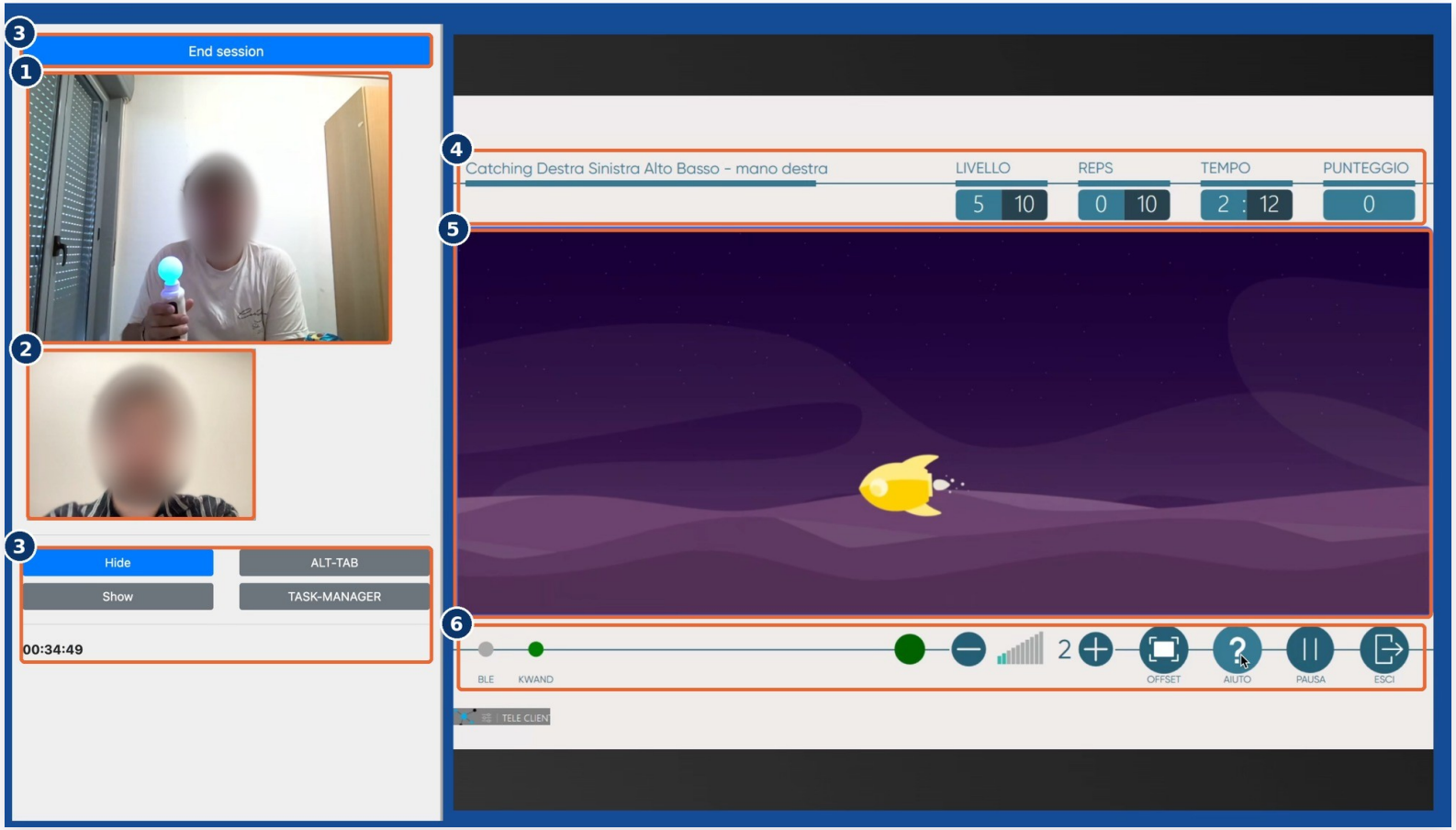
**50** **people took part**

46 volunteers · 4 patients

**34** **sessions analysed**

30 volunteers · 4 patients

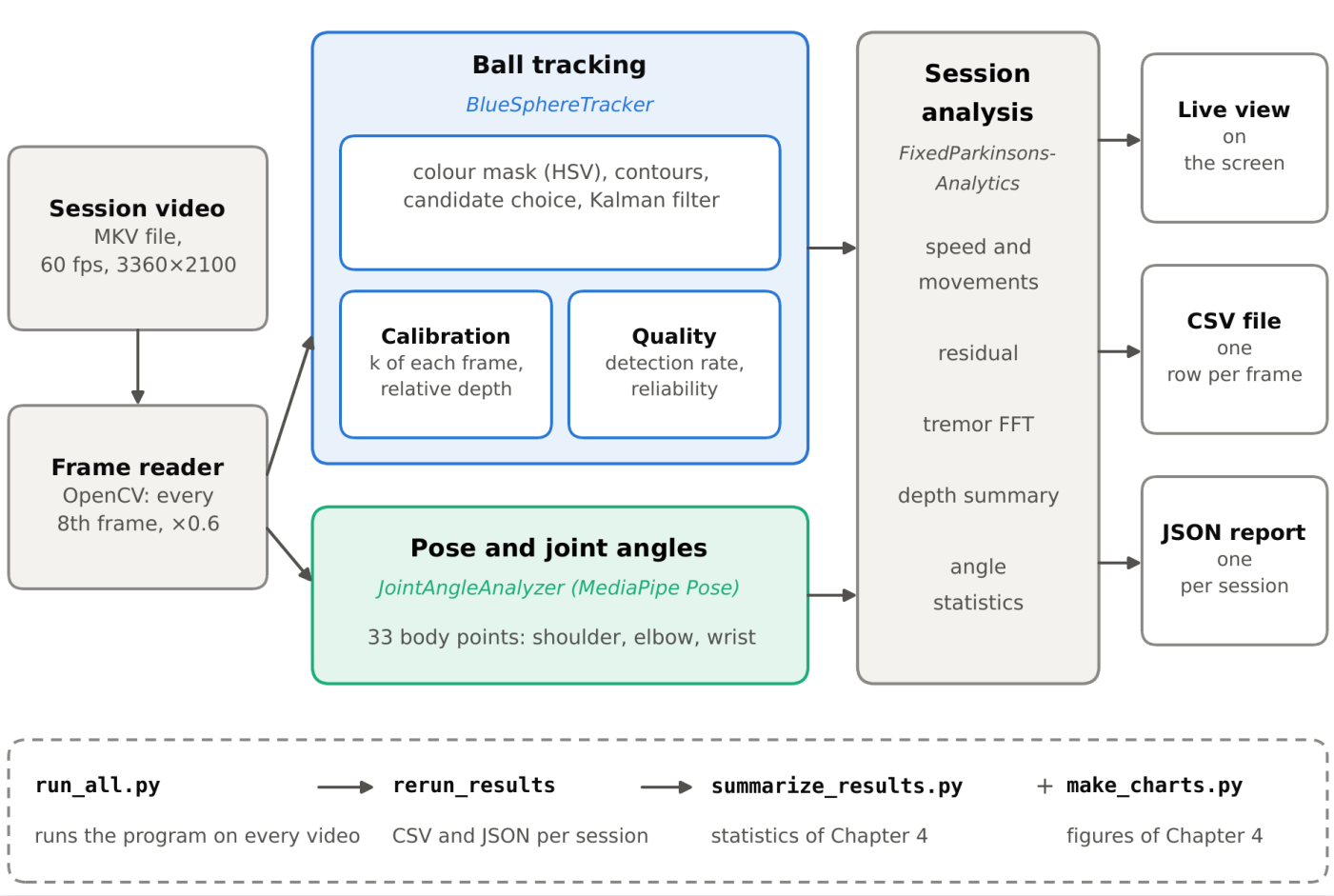
# What the physician sees today




*Faces blurred, browser address bar removed.*

1. Patient camera, with the K-Wand
2. Physician camera
3. Session controls and timer
4. Exercise: level, reps, time, score
5. Game area: the K-Wand moves the object
6. Device status and controls

# One video, two ways to measure




## Ball tracking

Follows the blue sphere: position, speed and movements in millimetres

## Pose estimation (AI)

MediaPipe finds the body: shoulder, elbow and wrist angles

## Outputs

A CSV file per frame, a JSON report per session and a live view

# Finding the sphere, frame by frame

1

## Blue pixels

HSV colour mask inside the tracking region

2

## Candidates

Blobs with the right size and roundness

3

## Best match

Nearest to the last position, smoothed by a Kalman filter

4

## Scale

Pixels to millimetres with the factor k

5

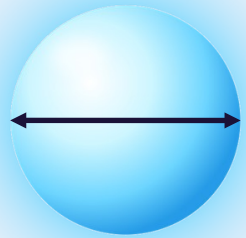
## Movements

Speed above a threshold marks one movement

Every 8th frame is processed: 7.5 positions per second

No candidate found: the Kalman filter predicts the position

# From pixels to millimetres



**d pixels**

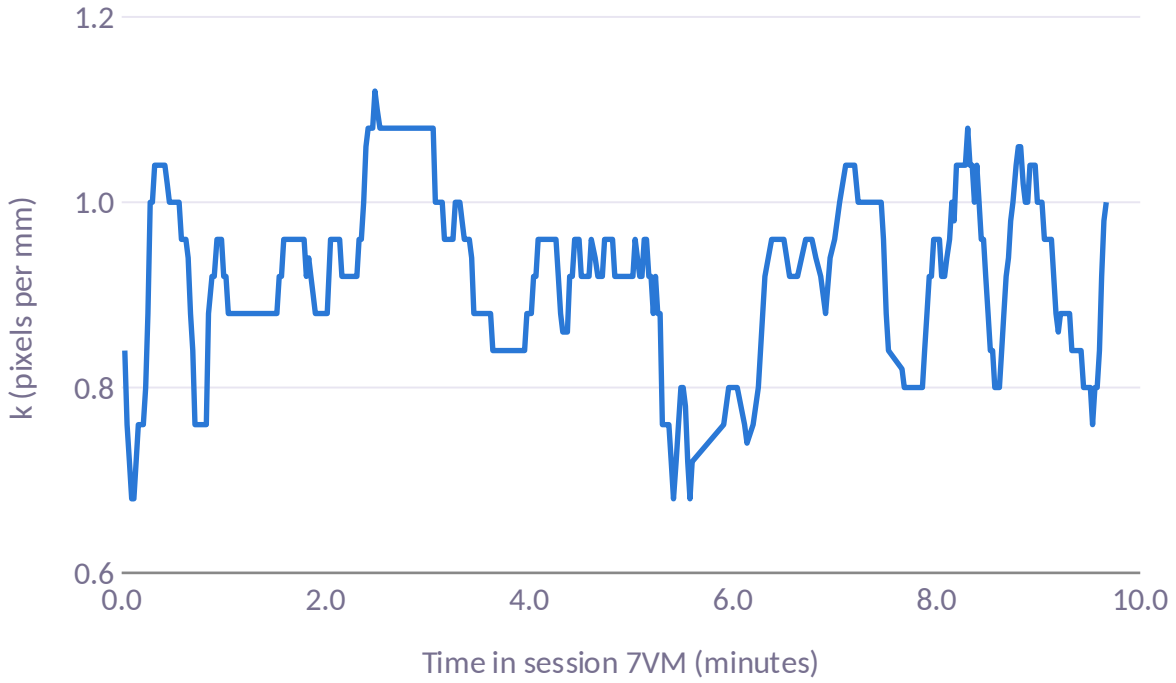
real diameter: 50 mm

## k = d / 50 mm

pixels per millimetre

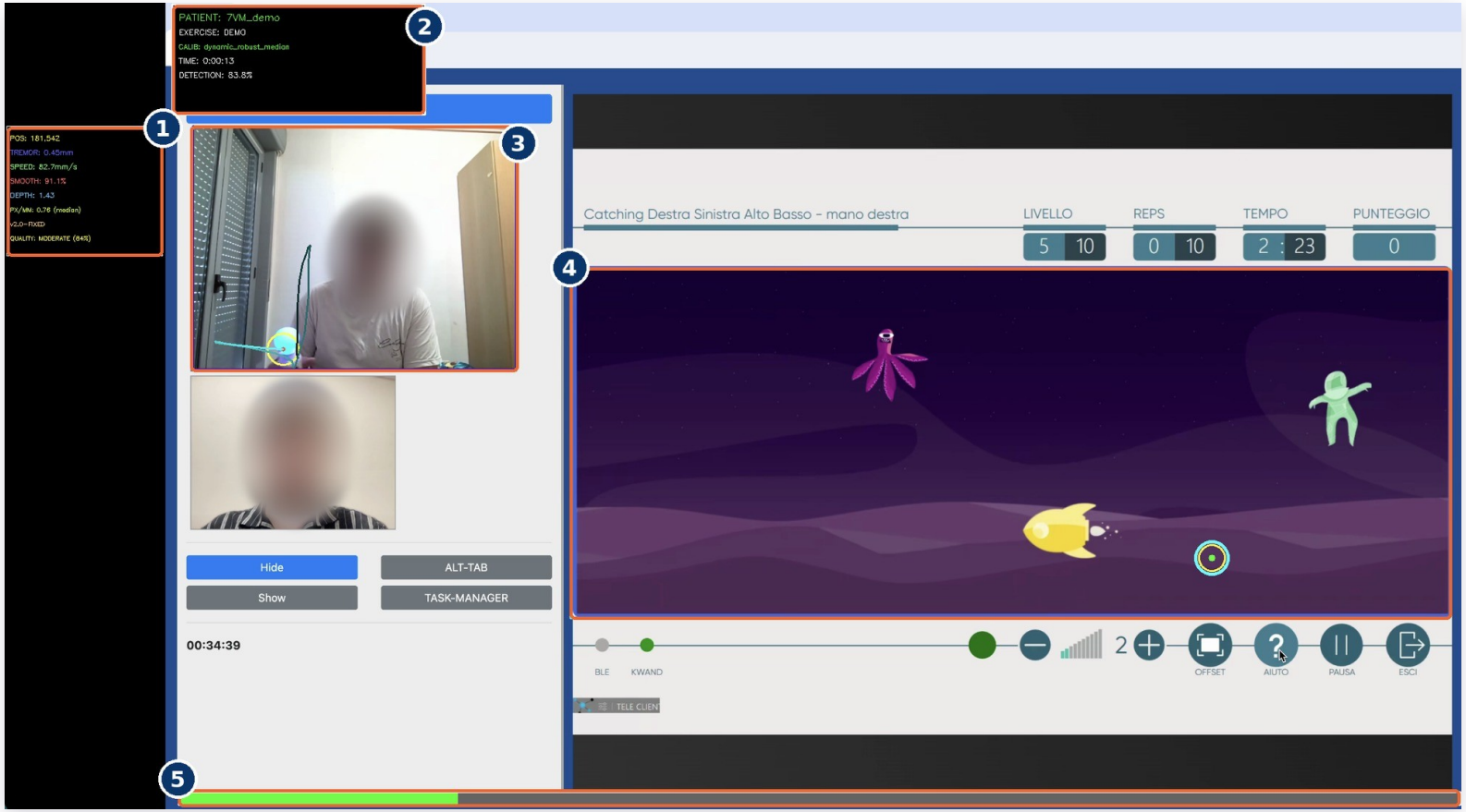
- k of each frame: median of the last 100 estimates
- Relative depth: $d_0$ / d (exploratory)

**The scale k during one session (7VM)**




Across the 34 sessions: k = 0.52 to 1.42 pixels per mm (median 0.95)

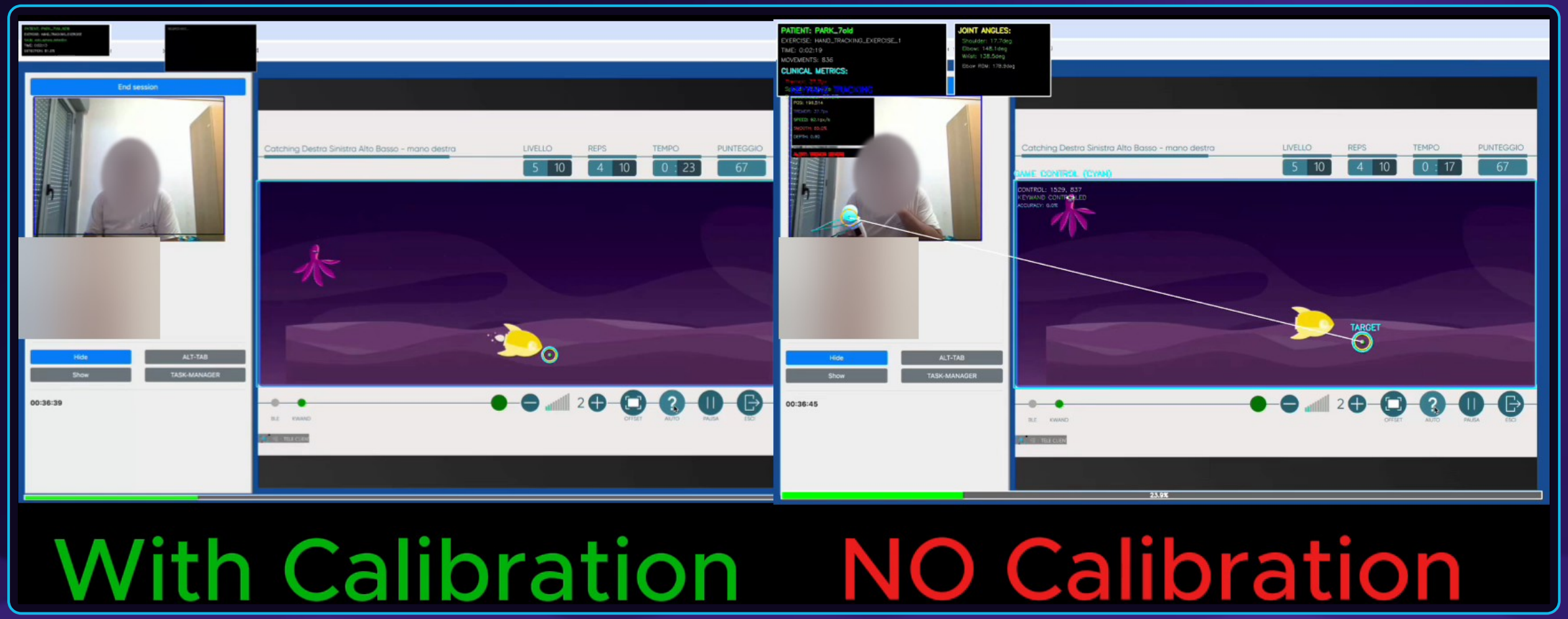
# What our system adds




*Faces blurred. The live tremor value is not used (see results).*

1. Live panel: position, speed, depth, k and quality
2. Session: participant code, time, detection rate
3. Tracking region: the sphere and its path
4. Game area: the detected position drawn as an object
5. Progress through the video

# With and without calibration




Left: with calibration (values in mm) · Right: without calibration (values in pixels) · faces blurred

# 03

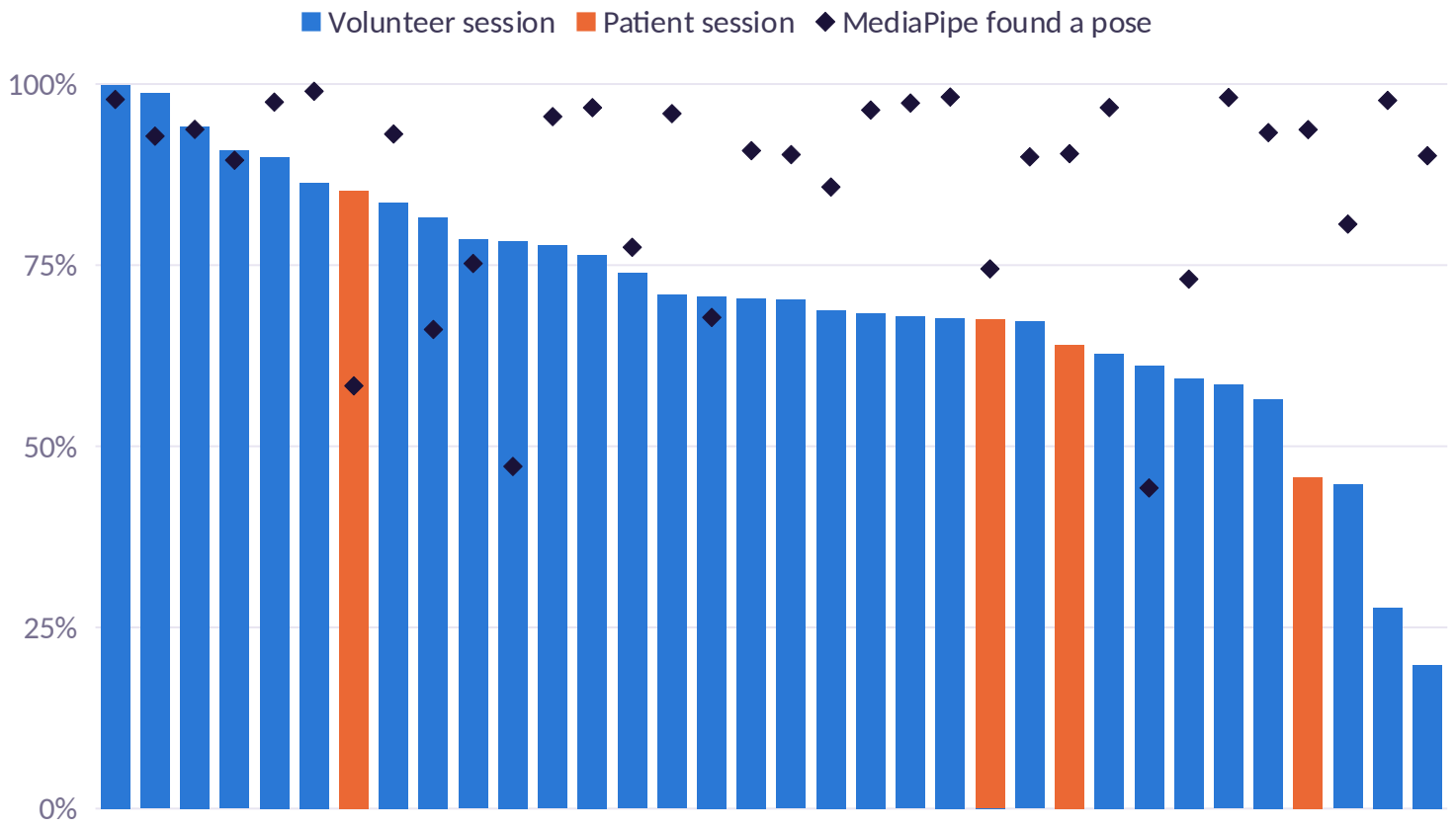
# Results

What the data shows: 34 sessions



# Fast enough for real time; light matters

**Share of frames with a detection, one bar per session (sorted)**




**0.58 ms**

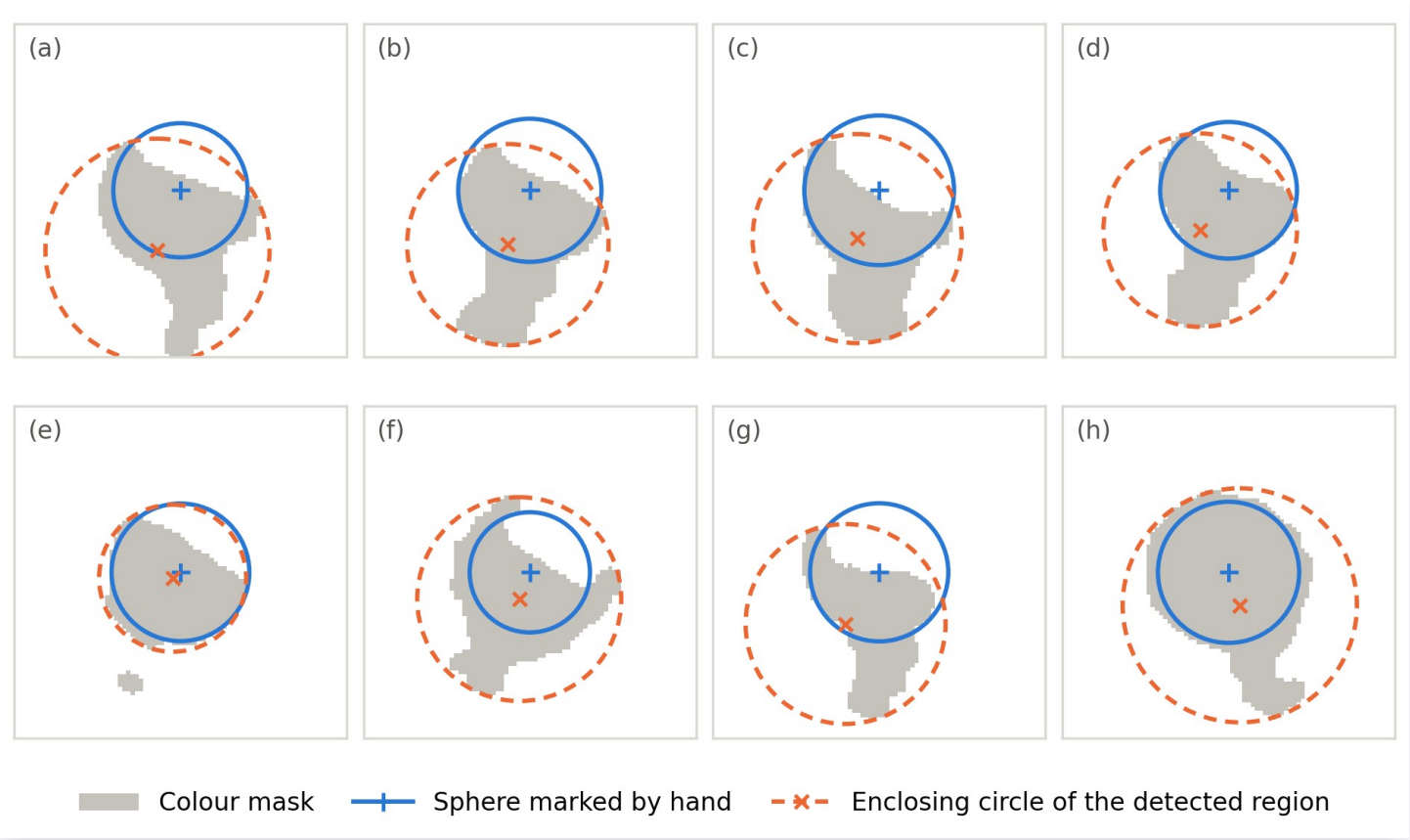
per frame for ball tracking (MediaPipe: 17.9 ms). Both are faster than a 30 fps webcam.

**20–100%**

of frames with a detection. It depends mainly on the lighting and the background.

*Detection rate is not accuracy: accuracy is on the next slide.*

# How accurate is the tracking?




*Eight of the 50 hand-checked frames of session 7VM*

**49 / 50**

hand-marked frames where the tracker followed the sphere

**≈ 21 mm**

median distance from the marked centre (13.5 px), mostly Kalman-filter lag

**≈ 30%**

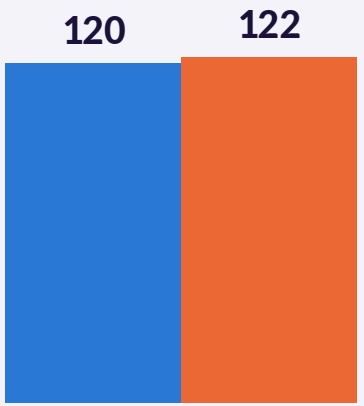
too small in mm: the mask misses the bright centre of the sphere

# Volunteers and patients: averages

Volunteers (30 sessions)  Patients (4 sessions)

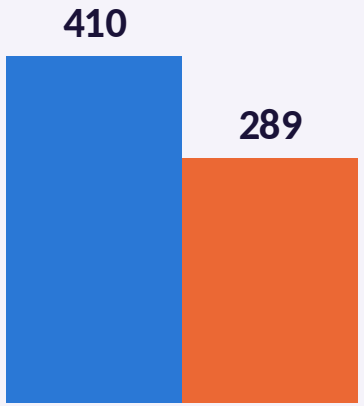
### Mean speed

mm/s



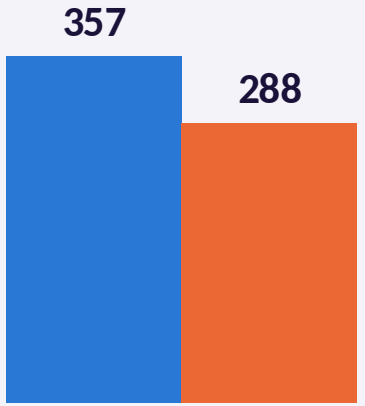

### Peak speed per movement

mm/s



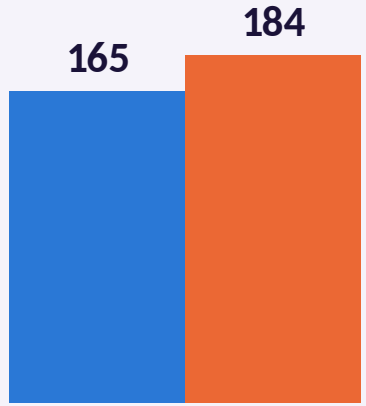

### Path per movement

mm




### Movements per session

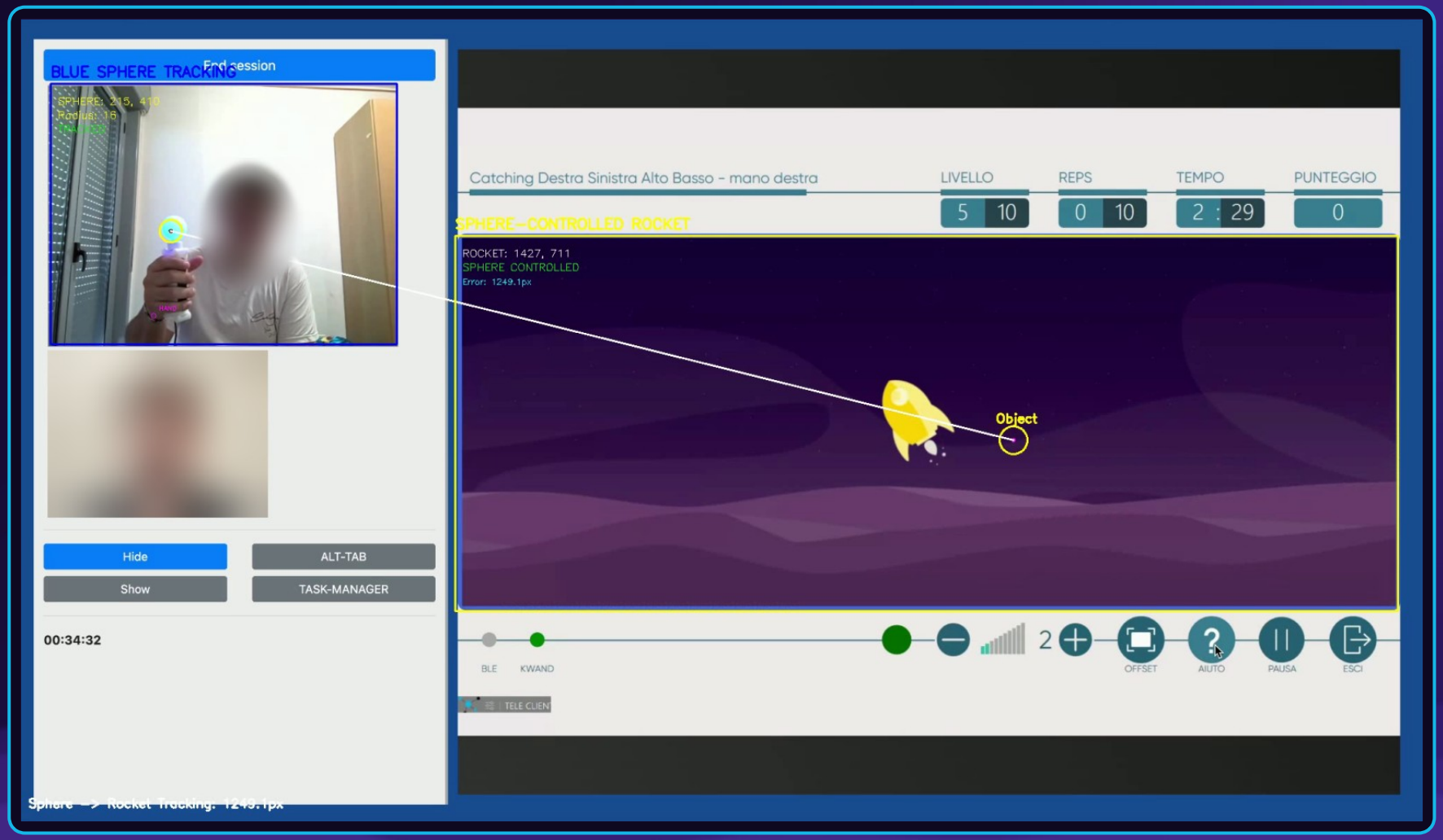
count




**Mean joint angles (relative values):** Shoulder 39° vs 73° · Elbow 98° vs 105° · Wrist 149° vs 142°

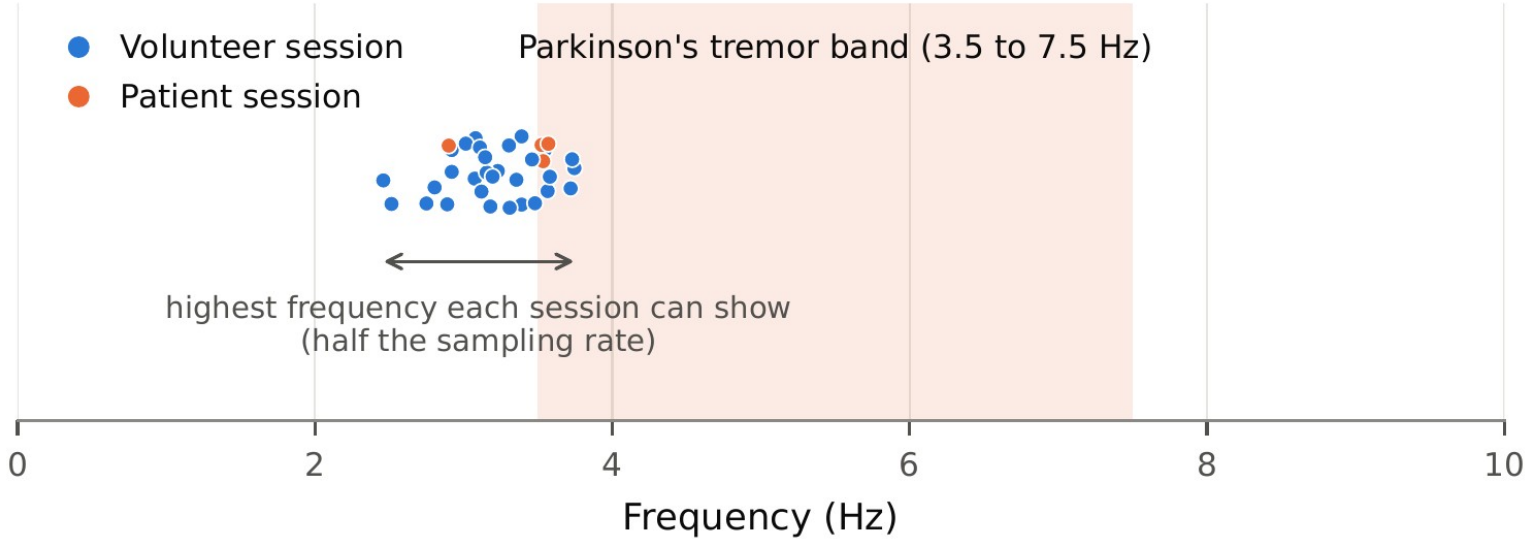
*With 4 patients these values describe the data; they are not a statistical test or a clinical finding.*

# A volunteer session




The system follows the sphere while the volunteer plays · faces blurred

# Why tremor cannot be measured here




## ≤ 3.7 Hz

the highest frequency we can see: half of 4.9 to 7.5 positions per second

## 3.5–7.5 Hz

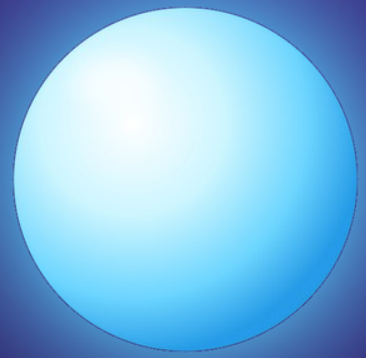
the tremor band of Parkinson's disease, above the limit in 25 of 34 sessions

**To measure tremor:** add rest periods to the exercise, use IMU sensors, or record at a higher sampling rate. Tremor is assessed at rest, not during active movement.

# 04

# Conclusions

What we learned and what comes next



# What this thesis shows

## Measures from video

A normal VRRS video gives movement measures in millimetres, with no extra sensors.

## Fast and complementary

Ball tracking runs in real time; MediaPipe adds the joint angles.

## Light is the main limit

Colour segmentation depends on the lighting and on the background.

## A feasibility study

With 4 patients the results describe the data; clinical validation comes next.

# Next steps: AI to improve detection

OBJECT DETECTOR

## YOLO

Finds the sphere without colour thresholds

SEGMENTATION

## U-Net

A precise sphere mask, robust to light

HAND LANDMARKS

## MediaPipe Hands

Tracks the hand directly: 21 points

MONOCULAR DEPTH

## Depth Anything

Distance from one camera

RECURRENT NETWORK

## LSTM

Learns movement patterns over time

**Clinical validation with more patients**

**Deployment: from the edge to a hospital-cloud dashboard**

# Thank you

Questions are welcome



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